Chemical Identity Lost in Regulation: A Study of Semantic Interoperability in European Chemical Substance Data

Abstract.

Chemical substance identity is inconsistently represented across European regulatory datasets, creating barriers to semantic interoperability. While structure-based identifiers such as InChI and InChIKey unambiguously define chemical entities, regulatory systems rely on heterogeneous identifiers (CAS numbers, EC numbers, textual names, and group descriptions) that frequently refer to mixtures, substance groups, or analytical parameters rather than defined molecular structures.

This study investigates the extent and consequences of this semantic mismatch and evaluates technical approaches for compensating or mitigating the resulting interoperability barriers.

We analyse 18 European regulatory datasets from ECHA and related sources (41,813 records total). We introduce a seven-tier linkability taxonomy to classify regulatory entries by degree of chemical specificity, assess the consistency of CAS number mappings, evaluate embedding-based and large-language-model semantic matching, estimate the effort required for manual reconciliation, and construct a knowledge-graph-based integration.

Only 62.5% of entries can be directly linked to a defined chemical structure; 37.5% are non-structure-based. CAS numbers exhibit both one-to-many and many-to-one mappings with molecular structures, undermining their role as unique identifiers. Embedding-based approaches fail to recover chemical identity (best model ChemOnt Hit@1 = 5.75%); Claude Sonnet 4.6 achieves Hit@1 = 39.2%, a sevenfold improvement, yet still misclassifies the majority of substances. Taxonomic alignment of legislative group entries with ChemOnt yields 304 validated SKOS mapping triples, illustrating what formal ontologies enable while confirming that post-hoc coverage remains bounded.

Semantic interoperability in chemical regulation cannot be achieved through post hoc technical solutions alone. It must be addressed at the level of data modelling and legislation, by adopting structure-based identifiers, formal ontologies, and machine-readable representations embedded in regulatory data standards. We provide scripts for constructing an integrated knowledge graph spanning the analysed regulatory sources.

Keywords. semantic interoperability, regulatory interoperability, Interoperable Europe, chemical substance identification

1. Introduction

Chemical substances are central entities in environmental, health, and product regulation. Across European legislation, substances are used to define obligations, assess risks, and communicate hazards in frameworks such as the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) [1], the Classification, Labelling and Packaging of chemicals (CLP) [2], and sector-specific regulations including the EU Pesticides Regulation [3] and the Water Framework Directive [4]. Governance of these substances increasingly depends on the ability to integrate and reuse data across regulatory domains.

In chemistry and cheminformatics, chemical identity is well-defined and operationalised through structure-based representations. Molecular structures can be uniquely encoded using standards such as SMILES notations [5], IUPAC names [6], the International Chemical Identifier (InChI) [7] and its hashed representation, the InChIKey, enabling unambiguous identification, comparison, and linking of substances across databases [8]. Large-scale infrastructures such as PubChem [9] and ChEBI [10] build on these identifiers to provide machine-readable, ontology-based knowledge graphs that support interoperability and automated reasoning.

In contrast, European regulatory data systems adopt a different approach. Substances are typically identified using heterogeneous and often ambiguous identifiers, including Chemical Abstracts Service (CAS) registry numbers [11,12], European Community numbers (EC-numbers) [13], textual names, and informal group descriptions. Moreover, the concept of a *substance* in regulation is not uniform: it may refer to a single molecule, a mixture of specific molecules, a mixture of unknown or variable composition (UVCB) [14], a group of related compounds (e.g., PFAS), or even an analytical parameter defined as a function over multiple substances. For a number of *substances*, the identity, Substance Identity Profile (SIP), analytical data about the composition of *UVCB substances*, usage information and toxicological information is not disclosed to the public in accordance to REACH Artikel 118 [15] and only a competent authority has access to complete REACH-dossiers. The general public and regional authorities have to rely on public ECHA info, CLP-classifications and safety data sheets (SDS) [16,17] which are not machine readable.

Figure 1 illustrates this contrast: the right panel shows the chemistry view, in which a substance corresponds to a molecule unambiguously identified by structure-derived identifiers such as InChI or InChIKey; the left panel shows the regulatory view, where the same term encompasses heterogeneous entities of different semantic types.

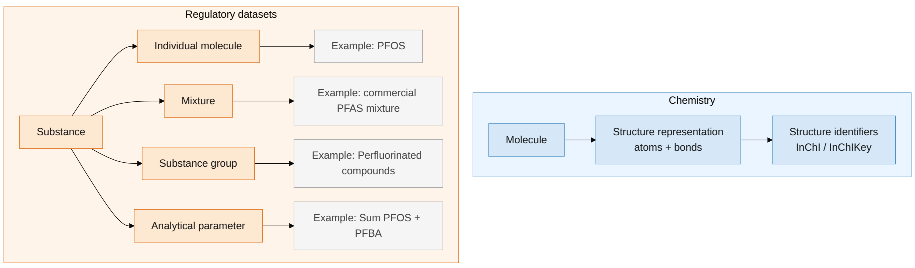

Figure 1. Conceptual model of the semantic mismatch between chemistry (right) and regulatory datasets (left). In chemistry, a substance corresponds to a molecule identified using structure-derived identifiers such as InChI or InChIKey. In regulatory datasets, the concept of a "substance" encompasses heterogeneous entities (individual molecules, mixtures, substance groups, and analytical sum parameters) that correspond to fundamentally different semantic types, complicating interoperability and automated reasoning. This mismatch is quantified empirically in Section 4.

As a result, integrating substance information across regulatory frameworks becomes a largely manual task, hindering policy enforcement for environmental administrations [18], and increasing uncertainty for industrial players in environmental management, investment, and innovation. Figure 2 illustrates this using PFAS as an example: individual molecules such as PFOS, PFBA, and PFBS are grouped into a regulatory substance group without explicit set-membership semantics, while an analytical parameter such as *Sum PFOS + PFBA* is a derived entity defined as a function over individual members. Relationships are not formally specified, leading to ambiguity and combinatorial complexity during cross-dataset harmonisation.

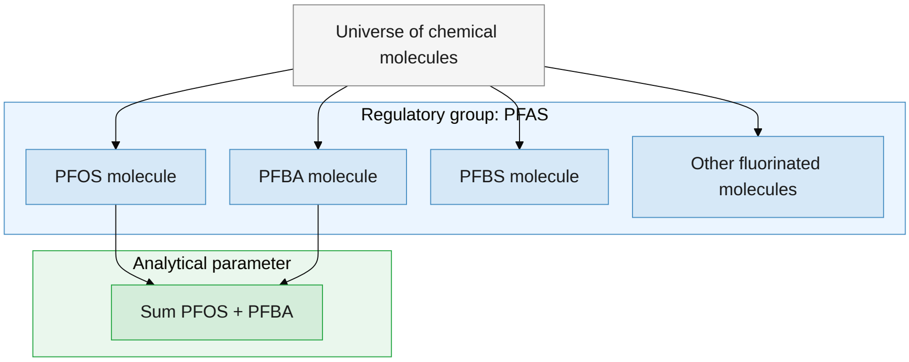

Figure 2. Set-theoretic view of a regulatory substance group (PFAS, upper subgraph) and a derived analytical parameter (lower subgraph). Individual molecules are grouped without explicit set-membership semantics, and the sum parameter aggregates contributions from named members.

This paper investigates the extent and consequences of this interoperability problem using an analysis of 18 regulatory datasets from the European Chemicals Agency (ECHA) and related sources. We focus on three main research questions:

RQ1 To what extent can substances in European regulatory datasets be linked to a unique chemical structure?
RQ2 How consistent and reliable are commonly used identifiers such as CAS numbers for cross-dataset linking?
RQ3 Can semantic, embedding-based or LLM methods compensate for the absence of structure-based chemical identifiers?

Based on this analysis, we make the following contributions:

C1 We quantify the heterogeneity of regulatory substance definitions and introduce a taxonomy of linkability reflecting their degree of chemical specificity.
C2 We demonstrate the limitations of CAS-based identification and show that a substantial fraction of regulatory substances cannot be resolved to a defined molecular structure.
C3 We evaluate semantic embedding and LLM approaches and show that they fail to recover chemical identity for non-structure substances.
C4 We estimate the scale of the manual effort required for retrospective harmonisation, demonstrating its infeasibility.
C5 We show that structure-based identifiers combined with ontology-driven approaches enable scalable integration through knowledge graphs.

The remainder of this paper is structured as follows. Section 2 summarises previous work. Section 3 describes the data sources and analytical methods. Section 4 presents the empirical findings. Section 5 discusses the implications for regulatory data infrastructures and semantic interoperability. Section 6 concludes with recommendations for future work.

2. Related Work

Interoperability of scientific data has been extensively studied in the context of FAIR principles [19] and knowledge graph infrastructures. In the life sciences, large-scale resources such as PubChem [9] and the Chemical Entities of Biological Interest (ChEBI) ontology [10] demonstrate how chemical entities can be represented using structure-based identifiers, ontology-driven classifications, and machine-readable knowledge graphs, enabling integration, querying, and reuse of chemical information across scientific, environmental, and regulatory domains. Rule-based chemical classification is further exemplified by ClassyFire [20], which establishes a structure-based taxonomy (ChemOnt) in which each class is characterised by structural rules.

Within the Semantic Web community, numerous approaches have been proposed for data integration based on linked data principles, ontology alignment, and entity resolution. Techniques such as identifier mapping, schema matching, and, more recently, embedding-based similarity methods [21] have been applied to address heterogeneity across datasets. However, these approaches typically assume that entities correspond to well-defined and comparable concepts.

In cheminformatics and regulatory science, approaches such as Kutsarova et al. [14] address individual substances without precise structural definitions using statistical sampling, but do not address broader semantic inconsistencies across regulatory datasets.

In the regulatory domain, previous work has highlighted challenges related to data fragmentation, identifier heterogeneity, and limited machine-readability of legal data [22]. Knowledge graph approaches such as the TOXIN knowledge graph [23] demonstrate that ontology-driven integration is applicable to regulatory chemical data, but address single-domain integration rather than cross-dataset semantic mismatch. The mismatch between chemical identity and regulatory substance definitions has not been systematically characterised; the present study addresses this gap.

3. Methods

Our analysis comprises the following steps:

Data Integration Substance records were compiled from 18 European regulatory and monitoring sources and one Flemish administrative list. These include ECHA obligation lists (Candidate List [24], Authorisation List [25], Restriction List [26], POPs List [27], EU Positive List [28], and the CLH Harmonised Classification and Labelling List [29]), ECHA activity lists (Restriction Process [30], SVHC Identification [31], Authorisation Process [32], Dossier Evaluation [33], CLH Process [34], Substance Evaluation [35], POPs Process [36], PBT Assessment [37], and ED Assessment [38]), as well as REACH registration data [39]. Additional sources comprise the EU Pesticides Database (Active

Substances) [40], the OSPAR List of Chemicals for Priority Action [41], and a Flemish administrative substance list developed in the context of water quality modelling [42]. This Flemish *summation substances* list reflects the heterogeneity of *substance names* in Flemish environmental permits and monitoring.

For each entry, available identifiers (CAS number, EC number, and substance name) were collected. Where possible, substances were linked to structure-based identifiers (InChIKey) via PubChem [9]. The resulting dataset establishes a unified mapping between regulatory entries, their identifiers, and their originating source lists.

Chemical Identity and Linkability Each substance entry was classified into one of several types reflecting its degree of chemical specificity: structure-defined molecule, substance group, mixture/UVCB, analytical parameter, or administrative entry.

Based on this classification, a seven-tier linkability taxonomy was defined, ranging from tier 1 (directly structure-resolvable via InChIKey) through partially resolvable entries (tiers 2–3), unresolved CAS numbers (tier 4), mixtures and UVCBs (tier 5), and administrative entries (tier 6), to tier 7 (no usable identifier). This taxonomy expresses the degree to which regulatory substances can be unambiguously linked across datasets.

Identifier Consistency Analysis For entries with both CAS numbers and InChIKeys, bidirectional mappings were analysed to assess identifier consistency. This analysis evaluates whether CAS numbers uniquely map to structures and vice versa.

Structure Coverage and Cross-List Overlap For each regulatory source, the proportion of structure-defined substances was calculated. Overlap between sources was analysed using structure-based matching (i.e., using InChIKey) and set similarity measures.

Analysis of Non-Structure Substances Substances without structure identifiers were analysed using text-based methods, including embedding-based clustering [43,44,45] and similarity matching to chemical taxonomies (i.e., using ChemOnt), to assess whether semantic techniques can recover meaningful structure or classification [46,20]. Each substance name and ChemOnt class label was encoded using the `all-MiniLM-L6-v2` sentence-transformer model [21,47,48], producing a 384-dimensional dense embedding vector per substance entry.

Embedding Model Validation Using ChemOnt Ground Truth To evaluate the ability of text-based embeddings to recover chemical classifications from substance names, an internal validation was performed using structure-defined substances as ground truth. Substances with an unambiguous InChIKey were linked to the ChemOnt taxonomy through ClassyFire classification [20], providing reference assignments at the levels of direct parent, subclass, class, superclass, and kingdom.

Substance names and ChemOnt class labels were encoded using pre-trained transformer models (`all-MiniLM-L6-v2`, `all-mpnet-base-v2` [49], `allenai/scibert_scivocab_uncased` [50], and `dmis-lab/biobert-v1.1` [51]) within the *sentence-transformers* framework [21,47,48]. Cosine similarity was used to rank all ChemOnt classes for each substance name.

Performance was assessed using Hit@k ($k = 1, 3, 5, 10$) [46]. To account for the hierarchical structure of ChemOnt, Hit@1 was additionally computed for each taxonomy level.

This validation provides a comparison of embedding models and demonstrates the extent to which semantic representations of substance names approximate structure-based chemical taxonomy assignments.

To assess whether large language models provide a more capable baseline, Claude Sonnet 4.6 was applied to the same validation benchmark under a zero-shot prompting protocol. For each substance name, the model was prompted to generate the ChemOnt class label at each hierarchy level (kingdom, superclass, class, subclass, and direct parent) together with a confidence estimate (high, medium, or low). Because the LLM generates a single predicted label per hierarchy level rather than ranking the full class space, only Hit@1 (accuracy) is reported; Hit@k for $k > 1$ is not defined for this setting. Performance metrics are therefore not directly comparable to the embedding retrieval results but address the same evaluation question.

LLM-Based Equivalence Assessment for Non-Structure Substances To identify regulatory group entries whose scope can be aligned with a ChemOnt taxonomy node, Claude Sonnet 4.6 was applied to a sample of non-structure substance names. For each entry, the model assessed whether the name describes a ChemOnt-equivalent group (rather than a single substance), identified the most specific equivalent ChemOnt node, and classified the relationship as `skos:exactMatch`, `skos:broadMatch` (regulatory entry is a subset of the ChemOnt class), or `skos:narrowMatch`. High-confidence assignments were serialised as SKOS triples and added to the knowledge graph.

Knowledge Graph Construction All data were integrated into an RDF knowledge graph using SKOS [52] for taxonomy representation. Structure-defined substances were linked to ChemOnt classes and to ChEBI entities via InChIKey, enabling integration of biological hazard annotations [53].

Workload Estimation For non-structure substances, interoperability requires manual assessment of pairwise relationships (equivalence, subset, overlap, disjoint). The required effort scales quadratically with the number of such entries as $\binom{N}{2} = N(N-1)/2$, where N is the number of non-structure entries under consideration.

Priority Scoring To illustrate the analytical value of the integrated knowledge graph (C5), a composite priority score was computed by combining regulatory list membership with biological role annotations from ChEBI [10], following the ToxPi framework [54]. This score is illustrative only and does not constitute a formal risk assessment.

4. Results

Overview of the Integrated Knowledge Graph We constructed an integrated knowledge graph to analyse the representation and interoperability of regulatory substances across datasets. Figure 3 provides an overview of the resulting graph structure and its main semantic layers. The graph is organised into four layers: an input layer of 18 regulatory sources; a core layer of ∼17,500 regulatory substance entities represented as SKOS concepts; linking layers connecting substances to structure-based identifiers (InChIKey), chemical classifications (ChemOnt), and biological roles (ChEBI); and derived layers comprising linkability tiers, embedding-based semantic clusters, and annotations (∼46,600 links).

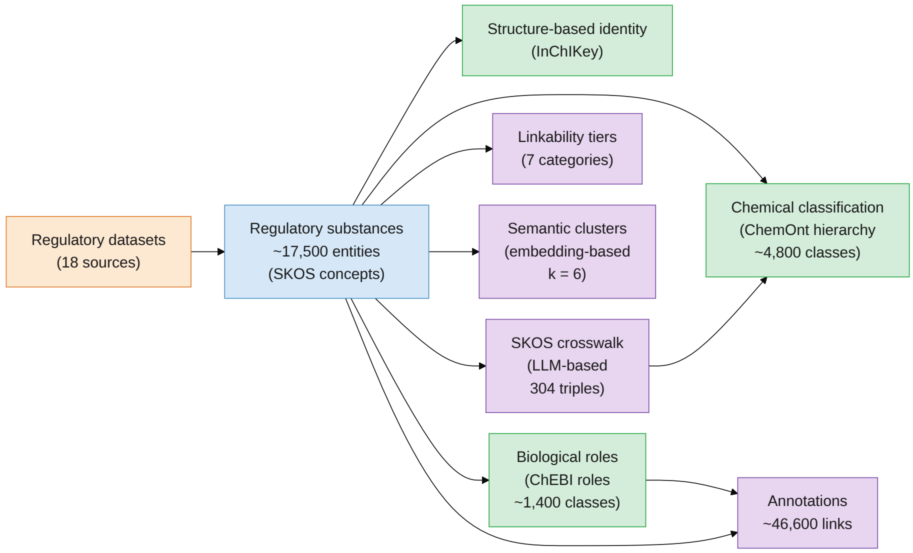

Figure 3. Overview of the integrated knowledge graph constructed from 18 regulatory datasets. The graph flows from the input layer of regulatory sources through the core layer of regulatory substances (SKOS concepts) to linking layers (InChIKey, ChemOnt, ChEBI) and derived layers comprising linkability tiers, semantic clusters, annotations, and an LLM-based SKOS crosswalk of 304 high-confidence triples linking non-structure substance groups to ChemOnt nodes.

Dataset Overview (C1) Integration of 18 regulatory sources yielded 41,813 records, of which 63% could be linked to a defined molecular structure via InChIKey.

Heterogeneity of Substance Definitions A large share of regulatory entries (37%) does not correspond to a defined chemical structure, consistent with Kutsarova et al. [14]. These include mixtures (UVCBs), substance groups, analytical parameters, or entries without usable identifiers, reflecting differences in how "substances" are defined across regulatory contexts.

Figure 4A shows that heterogeneity varies widely between regulatory lists: some sources consist almost entirely of structure-defined molecules, whilst others contain few structure-defined entries.

Applying the seven-tier linkability taxonomy (Section 3; Figure 4B) shows that 62.5% of entries (20,909) are directly linkable via InChIKey (tier 1). Only a small fraction consists of partially resolvable groups (tiers 2–3; <0.2%).

The remaining 37.5% cannot be directly linked to a structure: 11.6% have unresolved CAS numbers (tier 4), 10.8% are mixtures or UVCBs (tier 5), <0.1% are administrative entries (tier 6), and 14.9% lack any usable identifier (tier 7).

More than one-third of regulatory entries cannot be mapped to a unique chemical structure, and even CAS numbers do not guarantee unambiguous identification. Consequently, structure-based integration is limited to part of the data, and the non-structure fraction requires different analytical treatment.

Limitations of CAS Numbers (C2) CAS numbers are not reliable unique identifiers. Bidirectional mapping analysis reveals that 195 InChIKeys are associated with more than

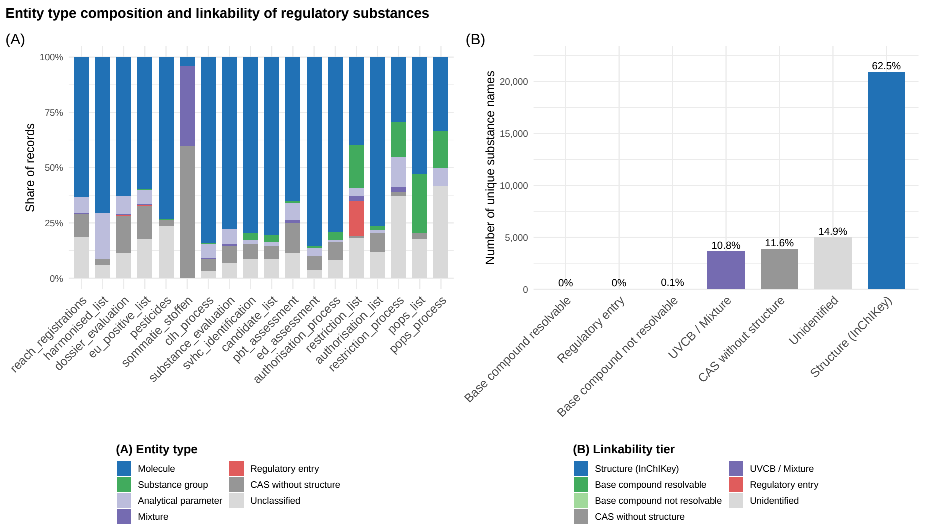

Figure 4. (A) Entity type composition across 18 regulatory sources; the mix of seven mutually exclusive types (Molecule, Substance group, Analytical parameter, Mixture, Regulatory entry, CAS without structure, Unclassified) differs substantially between regulatory instruments. (B) Linkability taxonomy applied to all 33,452 unique regulatory substance entries: 20,909 entries (62.5%) are directly linked via InChIKey (tier 1); 4,981 (14.9%) lack any usable identifier. Together, the panels show that structural linkability is unevenly distributed and that more than one-third of all entries cannot be resolved to a unique chemical structure.

one CAS number (up to 7 per structure), while 4 CAS numbers resolve to two distinct InChIKeys. These inconsistencies arise from differences in stereochemistry, salt forms, and context-dependent registration practices [8,55]. CAS-based cross-dataset linking therefore introduces ambiguity rather than resolving it, making CAS numbers insufficient as a backbone for interoperability.

Limited Interoperability Across Regulatory Lists Cross-list analysis of the 17,523 structure-resolved InChIKeys shows that 70.0% (12,270 entries) appear in only a single regulatory source, while just 30.0% (5,253 entries) are shared across two or more lists. This limited overlap reflects the domain-specific design of regulatory instruments rather than a shared chemical reference framework.

Failure of Semantic Approaches for Non-Structure Substances (C3) Embedding-based clustering of non-structure substance names yields semantically coherent groups but does not resolve chemical identity. While clusters reflect lexical or contextual similarity among substance descriptions, they do not correspond to well-defined molecular entities (Figure 5A), confirming that neither lexical context nor pre-trained language models can recover chemical structure from regulatory substance names.

To evaluate whether semantic similarity can recover chemical meaning, cosine similarity between sentence embeddings of substance names and ChemOnt class labels was computed for four models (`all-MiniLM-L6-v2`, `all-mpnet-base-v2`, `allenai/scibert\_scivocab\_uncased`, `dmis-lab/biobert-v1.1`). Score distributions are heavily skewed towards lower values and are not comparable across models, as embedding geometry differs between architectures (Figure 5B); a fixed threshold is therefore model-dependent and not a principled quality criterion. Manual inspection of the top-

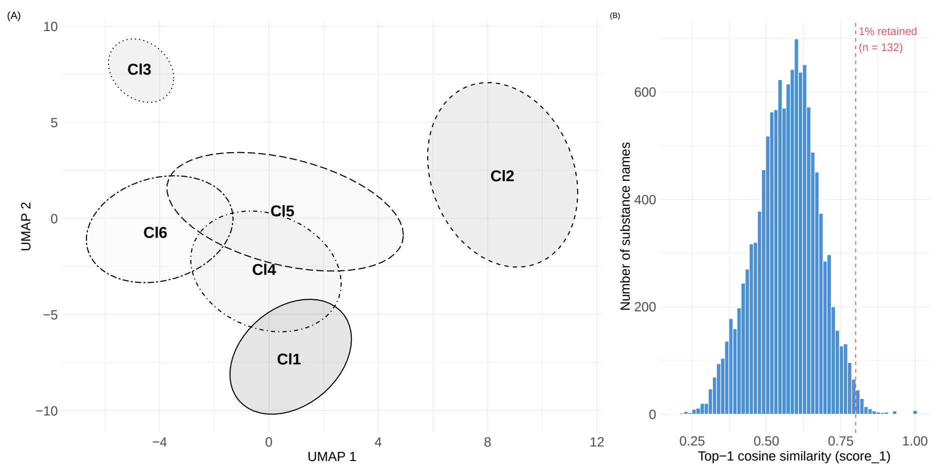

Figure 5. (A) UMAP projection of sentence embeddings for 12,499 non-structure regulatory substance entries, grouped by k-means cluster (k = 6); spatial proximity reflects semantic similarity in the 384-dimensional embedding space. Clusters: Cl1 = Reaction masses; Cl2 = Complex substances (UVCB with structure); Cl3 = Petroleum & coal tar fractions; Cl4 = Inorganic compounds & metal salts; Cl5 = Trade names, codes & biological materials; Cl6 = Polymers, fatty acids & surfactants. (B) Distribution of top-1 cosine similarity scores between substance name embeddings and ChemOnt class label embeddings for `all-MiniLM-L6-v2`; the dashed line indicates the threshold ($s_1 > 0.80$) used to bound manual evaluation (132 entries, 1.1%).

200 matches per model confirmed that the large majority of high-scoring assignments are chemically incorrect; correct cases were limited to surface-form overlap between substance name and class label (e.g. *Phenols* → *Phenols*). Score magnitude is not a reliable indicator of match correctness: models with the highest score floors (SciBERT, BioBERT) performed worst on both manual inspection and quantitative validation.

To validate this failure quantitatively, the four models were evaluated against a structure-derived ChemOnt ground truth comprising 28,204 ClassyFire-assigned substances. These results, shown in Table 1, indicate that `all-MiniLM-L6-v2` achieved the highest Hit@1 of 5.75% — even this best result implies the correct class is retrieved in fewer than 1 in 17 attempts. All other models scored below Hit@1 = 1%. Retrieval performance decreased further with increasing taxonomic abstraction: Hit@1 fell to 3.25% at the subclass level and below 1% for class, superclass, and kingdom.

Table 1. Performance of embedding models and LLM for ChemOnt classification.

Model	Hit@1	Hit@3	Hit@5	Hit@10
all-MiniLM-L6-v2	0.0575	0.0986	0.1234	0.1621
all-mpnet-base-v2	0.0093	0.0152	0.0184	0.0247
SciBERT	0.0040	0.0071	0.0102	0.0131
BioBERT	0.0047	0.0077	0.0099	0.0150
Claude Sonnet 4.6[†]	0.392	—	—	—

[†]LLM generates a single label per hierarchy level, not a ranking over the full class space.

Large language model classification substantially outperforms embedding-based retrieval: Claude Sonnet 4.6 achieves a Hit@1 of 39.2%, a seven-fold improvement over the best embedding model. Performance is monotonically higher at more general levels

(96.5% at kingdom, 52.3% at superclass), and LLM-assigned confidence is a validated quality signal: high-confidence assignments reach 53.9% accuracy, while low-confidence cases score 0%. Nevertheless, even at its strongest, the LLM correctly classifies fewer than 2 in 5 substances at the most specific level, and application to non-structure substances yields substantially higher null rates and lower confidence, confirming that linguistic structure alone cannot substitute for chemical structure information.

Scale of the Interoperability Problem (C4) For the more than 11,000 non-structure entries, full manual reconciliation would require on the order of hundreds of millions of pairwise comparisons. Under conservative assumptions, this corresponds to several thousand person-years of expert effort.

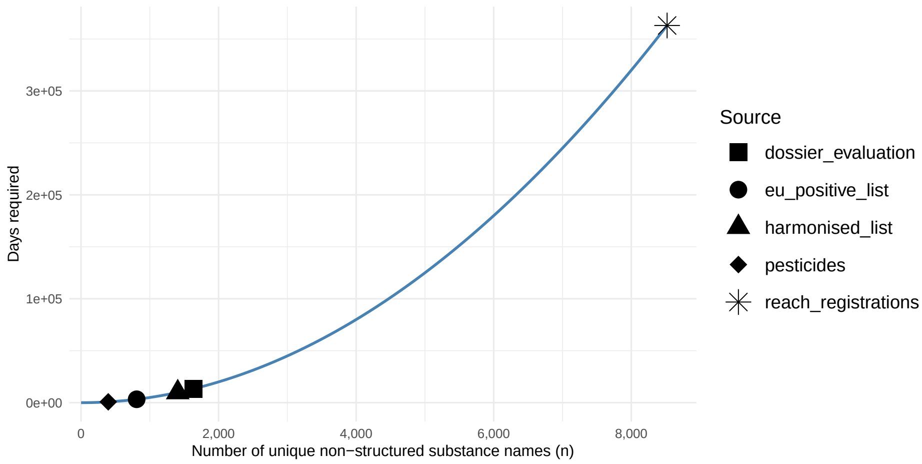

Figure 6. Estimated pairwise curation workload for non-structure entries across the combined European regulatory dataset, broken down by source. Reach registrations correspond to approximately 360,000 person–days (1,800 person-years) under an assumed throughput of 100 pair assessments per person-day. The quadratic growth of the workload illustrates that manual reconciliation is not scalable, reinforcing the need for structure-based and ontology-driven representations at the source.

Figure 6 illustrates this quadratic growth across all regulatory sources; the workload is dominated by REACH registrations (1,800 person-years alone), and each additional non-structure entry is disproportionately costly.

Knowledge Graph Integration and Priority Scoring (C5) Structure-defined substances can be successfully integrated across regulatory lists, chemical taxonomies (ChemOnt), and biological annotations (ChEBI) using InChIKey as a linking key.

This enables cross-domain queries that are infeasible on fragmented regulatory datasets: for example, identifying substances that appear on three or more ECHA lists and carry a documented biological role in ChEBI, or retrieving all members of a given ChemOnt class that are subject to authorisation requirements. As an illustration of this analytical potential, a composite priority score combining regulatory list membership with ChEBI biological role annotations was computed following the ToxPi framework [54]. Figure 7 plots regulatory versus biological scores for all scored substances; 2,2-bis(4-hydroxyphenyl)propane and dibutyl phthalate exemplify substances with both broad regulatory presence and multiple documented biological roles. This score is illus-

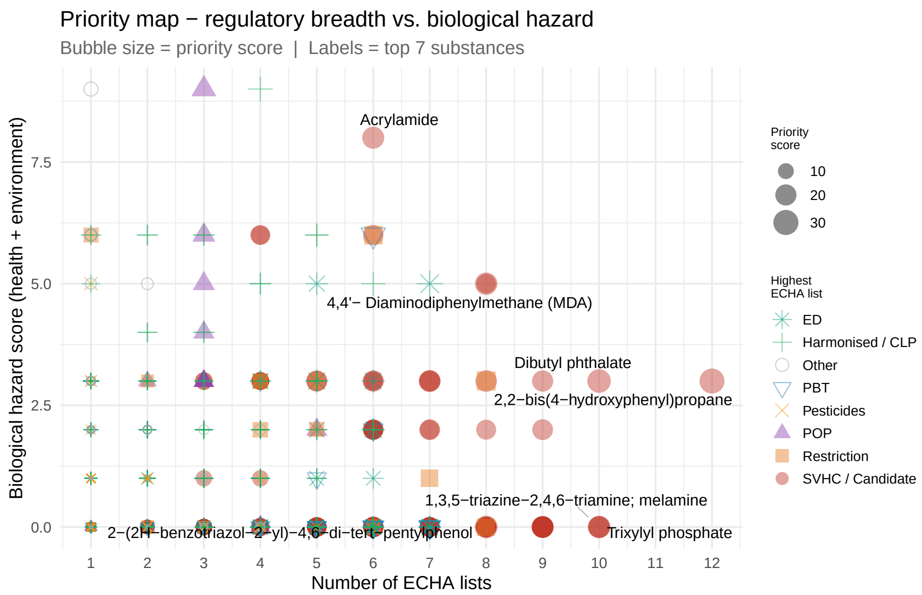

Figure 7. Bubble chart of composite priority space: regulatory list score versus biological hazard score for all scored substances. Bubble size is proportional to the number of regulatory source memberships. Substances in the upper-right quadrant combine broad regulatory coverage with documented biological hazard evidence.

trative and does not constitute a formal risk assessment. Extending the knowledge graph to non-structure substance groups, LLM-based equivalence assessment identifies 31.7% of sampled non-structure entries as ChemOnt-equivalent group entries—far above the 0.8% detected by regular-expression pattern matching. Of these, 304 high-confidence SKOS mapping triples were validated (285 `skos:broadMatch`, 18 `skos:exactMatch`, 1 `skos:narrowMatch`), forming a directly reusable legislative–ChemOnt crosswalk. The dominance of `skos:broadMatch` (92.8%) reflects that regulatory group entries typically define constrained subsets of ChemOnt classes (e.g. *fatty acids, C16–22* $\subset$ *Fatty acids and conjugates*), making their scope explicit and their member substances enumerable via the ChemOnt hierarchy. The remaining 68% of non-structure entries could not be reliably mapped, underscoring the limits of post-hoc taxonomic alignment.

5. Implications

Structural Linkability and Its Limits (RQ1) The results demonstrate that interoperability of regulatory substance data is constrained by the absence of a shared and formalised notion of chemical identity. While structure-based identifiers such as InChIKey enable unambiguous linking across datasets, a substantial fraction of regulatory entries cannot be expressed in this form. Instead, regulatory "substances" frequently represent heterogeneous entities, including mixtures, substance groups, and analytical parameters. This reflects a structural mismatch between chemical reality and regulatory abstraction: without explicit formalisation of equivalence, subset, and overlap relations, automated integration remains limited.

Unreliability of CAS-based Identification (RQ2) The analysis confirms that commonly used identifiers in regulatory contexts, particularly CAS numbers, do not provide a reliable foundation for interoperability. Although widely adopted, they exhibit both one-to-many and many-to-one mappings with chemical structures, reflecting differences in stereochemistry, salt forms, and context-dependent definitions. This finding challenges the widespread assumption in regulatory data integration that such identifiers can serve as globally unique references. In practice, relying on them introduces ambiguity rather than resolving it, undermining their suitability as a primary reference mechanism for cross-dataset linking.

Failure of Semantic Compensation (RQ3) Embedding-based methods and large language models cannot compensate for the absence of structure-based identifiers. The limitations are inherent to the data: textual descriptions of mixtures, groups, and analytical entries lack the information required to infer molecular structure, so semantic similarity reflects lexical overlap rather than chemical identity. Advanced tooling raises the ceiling but does not dissolve this constraint.

The Combinatorial Barrier to Retrospective Harmonisation The scale of the interoperability problem becomes evident when considering the effort required for manual reconciliation. For non-structure substances, establishing semantic relationships requires pairwise comparison, leading to quadratic growth in workload. The estimated effort of several thousand person-years demonstrates that retrospective correction is not a viable strategy at scale, interoperability must be built into regulatory data models by design.

Towards Ontology-Driven Regulatory Data Infrastructures The 304 high-confidence SKOS triples illustrate concretely what ontology alignment enables: via `skos:broadMatch` ($n = 285$), a regulatory description such as *fatty acids, C16–22* acquires machine-interpretable scope and its candidate member substances become enumerable through the ChemOnt hierarchy; via `skos:exactMatch` ($n = 18$), the ChemOnt class directly and completely defines the regulated group. For structure-defined substances, InChIKey-based linking and ChEBI annotations further enable cross-domain queries and composite scoring that are infeasible on fragmented datasets. However, the crosswalk covers only entries whose names contain sufficient chemical information, highlighting that the interoperability benefit of ontologies is maximised when group boundaries are defined explicitly in the regulatory text itself.

Implications for FAIR Chemical Data in Europe The findings have direct implications for the implementation of FAIR (Findable, Accessible, Interoperable, Reusable) principles in European chemical regulation. Current regulatory datasets fall short of interoperability primarily because they lack globally unique, persistent, and machine-readable identifiers grounded in chemical structure. Achieving FAIR chemical substance data requires a shift from document-centric and identifier-fragmented approaches towards linked data infrastructures [56,57] in which substances are represented as resolvable URIs, grounded in structure-based identifiers and connected through formal ontologies. This shift is not purely technical: it requires alignment between data modelling practices and the legal definitions of substances in regulatory frameworks.

Limitations and Scope This study focuses on datasets derived from ECHA and related European regulatory sources, and on the use of InChIKey as the primary structure-based identifier. While the results are consistent across the analysed datasets, further work is needed to assess the generality of these findings across other regulatory domains, jurisdictions, and substance categories. Furthermore, not all regulatory concepts can or should be reduced to individual molecular structures. Mixtures, UVCBs, and analytical parameters are legitimate and necessary constructs in regulation. However, their current representation lacks formal semantics, limiting their integration with structure-based systems. Future work should explore hybrid models that explicitly represent these entities as sets, classes, or functions over well-defined chemical structures.

From Data Integration to Regulatory Design The interoperability problem is not primarily a technical integration challenge, but a consequence of how substances are defined and represented in regulation. Efforts to improve interoperability must therefore extend beyond data engineering to include the design of regulatory concepts, vocabularies, and data standards. Embedding structure-based identifiers, ontology references, and machine-readable semantics directly into regulatory processes [58,18] would reduce ambiguity and enable scalable integration. Without such changes, the gap between regulatory data and scientific data infrastructures will continue to widen, limiting the effectiveness of data-driven environmental governance. In practice, this implies that regulatory data infrastructures should: (i) include structure-based identifiers (e.g. InChIKey) wherever applicable; (ii) represent substance groups and mixtures using explicit semantic models (e.g. set- or class-based representations); (iii) publish substance data as linked data with persistent URIs.

6. Conclusion

In this study, we show that interoperability of chemical substance data in European regulatory frameworks is limited by the lack of a shared and formalised notion of chemical identity. Analysis of 18 regulatory datasets reveals that a substantial fraction of substances cannot be unambiguously linked to a molecular structure, and that commonly used identifiers such as CAS numbers are insufficient for reliable and consistent cross-dataset integration.

The three research questions are answered as follows. RQ1: A seven-tier linkability taxonomy (Section 3) applied to 41,813 records shows that 62.5% of regulatory entries (20,909) resolve directly via InChIKey (tier 1), while the remaining 37.5% cannot be linked to a defined chemical structure: 11.6% carry unresolvable CAS numbers (tier 4), 10.8% are mixtures or UVCBs (tier 5), and 14.9% lack any usable identifier (tier 7). RQ2: CAS numbers prove unreliable as unique identifiers; bidirectional mapping shows that 195 InChIKeys are each linked to more than one CAS number (some carrying up to seven) and 4 CAS numbers resolve to two distinct InChIKeys, with inconsistencies arising from differences in stereochemistry, salt forms, and registration context. Cross-list analysis further shows that 70% of structure-resolved InChIKeys appear in only a single regulatory source, reflecting domain-specific rather than shared chemical reference frameworks. RQ3: Semantic and embedding-based approaches cannot compensate for the absence of structure-based identifiers; the best-performing model achieves a ChemOnt Hit@1 of only 5.75%, meaning the correct chemical class is retrieved in fewer than 1 in 17 attempts, and all other models scored below Hit@1 = 1%. Applying Claude Sonnet 4.6 to the same benchmark yields Hit@1 = 39.2%, a seven-fold improvement, yet still classifying fewer than 2 in 5 substances correctly at the most specific level; confidence-filtered subsets reach 53.9% accuracy but cover only a minority of entries. Manual inspection reveals that high cosine similarity scores reflect surface-form lexical overlap rather than chemical understanding, a failure inherent to the data: textual descriptions of mixtures, substance groups, and analytical parameters simply do not contain sufficient information to infer molecular structure.

These limitations directly affect semantic interoperability. Many regulatory definitions lack explicit structure-based identifiers and formal semantics, leaving relationships such as equivalence, inclusion, or overlap implicit. This is particularly evident for substance groups and analytical parameters, which often represent sets or transformations of underlying chemical entities (Section 4). Without explicit formalisation of these relationships, interoperability and automated reasoning remain constrained.

The scale of the problem becomes clear when considering the effort required for manual reconciliation. For the more than 11,000 non-structure entries, establishing semantic relationships requires pairwise comparison, leading to quadratic growth in curation workload. Under conservative assumptions this corresponds to approximately 630,000 person-days (3,150 person-years) of expert effort, demonstrating that retrospective harmonisation is not operationally feasible. In contrast, structure-defined substances can be successfully integrated at scale: the knowledge graph constructed in this study (Section 4) links regulatory entities to InChIKey, ChemOnt, and ChEBI annotations across approximately 46,600 cross-domain links, extended by a legislative–ChemOnt crosswalk of 304 high-confidence SKOS triples making previously implicit group boundaries machine-interpretable: for the 303 `skos:broadMatch` and `skos:exactMatch` entries, member substances become directly enumerable via the ChemOnt hierarchy—enabling cross-domain queries and composite priority scoring (Section 3) that are infeasible on fragmented regulatory datasets.

Our results indicate that interoperability cannot be achieved through post hoc technical solutions alone. Instead, it requires changes at the level of data modelling and regulation: the inclusion of structure-based identifiers (e.g. InChIKeys), semantic models for substance groups and mixtures, and publication of regulatory data as linked data with persistent, resolvable URIs. Without such changes, fragmentation between regulatory and scientific data infrastructures will persist.

Future work should address several open directions. First, formal semantic representations are needed for mixtures, UVCBs, and substance groups that can express their compositional and set-theoretic structure in a machine-readable way, for example representing a substance group as an explicit set of member InChIKeys, or an analytical sum parameter as a function over named constituents. Second, these representations must be translated into operational standards for regulatory data exchange, embedding interoperability requirements directly into the design of future regulatory instruments and reporting obligations. Third, the generality of the findings should be assessed beyond European ECHA datasets, across other regulatory domains, jurisdictions (e.g. US EPA and UK HSE), and substance categories where similar semantic mismatches are likely present. Fourth, the 37.5% of non-structure entries that currently resist automated linking remain an open challenge: curated resolution workflows, possibly guided by the linkability taxonomy introduced here, could provide a starting point for targeted harmonisation. Finally, the knowledge graph infrastructure developed as part of this study provides a foundation for further work on automated reasoning, regulatory impact assessment, and cross-domain prioritisation, capabilities that depend critically on the continued expansion of structure-grounded and ontology-linked regulatory data.

Data Availability

All data, code, and materials supporting this study are publicly available. The complete dataset, including the integrated knowledge graph, analysis scripts, and figures, is archived on Zenodo at `https://doi.org/10.5281/zenodo.xxxxxxx`[1].

Supplementary Materials

The integrated knowledge graph (RDF/Turtle), processed datasets from 18 regulatory sources, analysis scripts, and all figures are available via the Zenodo archive associated with this publication.

Declaration on the Use of Generative AI

Research subject. Claude Sonnet 4.6 (`claude-sonnet-4-6`, Anthropic) was used as the *object of study* in three analyses reported in this paper: LLM-based ChemOnt classification benchmarking (to explore whether an LLM can compensate for the absence of structure-based identifiers) and LLM-based equivalence assessment for SKOS mapping of legislative group entries to ChemOnt nodes (to explore whether a legislative–ChemOnt crosswalk is constructible). The model was queried via the Anthropic API with prompt caching enabled, consuming \$100 tokens.

Development tool. Claude Code (Anthropic, model: `claude-sonnet-4-6`) was used during the preparation of this work for: (1) writing, debugging, and documenting analysis code in R, Bash, and Python; and (2) assisting with system configuration to enable local execution of transformer-based embedding models. These locally executed embedding models consumed negligible additional energy relative to the LLM API experiments. After using this tool, the author reviewed and edited all content as needed and takes full responsibility for the publication's content.

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